WILSON, M. 1970. Banff National Park archaeological survey 1969: faunal remains. Department of Archaeology, University of Calgary, Calgary, Alta., unpublished report on file, 35 p.
——— 1973. The early historic fauna of southern Alberta: some steps to interpretation. *In* Historical archaeology in northwestern North America. *Edited by* R. M. Getty and K. R. Fladmark. University of Calgary, Calgary, Alta., pp. 213–248.
——— 1980. Morphological dating of late Quaternary bison on the northern plains. Canadian Journal of Anthropology, **1**(1), pp. 81–85.
——— 1983. Canid scavengers and butchering patterns: evidence from a 3600-year-old bison kill site. *In* Carnivores, human scavengers and predators: a question of bone technology. *Edited by* G. M. LeMoine and A. S. MacEachern. Proceedings, 15th Annual Conference, Chacmool, The Archaeological Association of the University of Calgary, Calgary, Alta., pp. 95–139.
WISHART, W. D. In press. Western Canada. *In* White-tailed deer: ecology and management. *Edited by* L. K. Halls. Stackpole Co., Harrisburg, PA.
YOUNG, S. P. 1956. The deer, the Indians and the American pioneers. *In* The deer of North America. *Edited by* W. P. Taylor. Stackpole Co., Harrisburg, PA, pp. 1–27.

Paleomagnetism of the Mudcracked formation of the Precambrian Little Dal Group, Mackenzie Mountains, Northwest Territories, Canada[1]

JOHN K. PARK
Division of Seismology and Geomagnetism, Earth Physics Branch, 1 Observatory Crescent, Ottawa, Ont., Canada K1A 0Y3

Received July 15, 1983
Revision accepted November 15, 1983

A previous study detailed the paleomagnetism of red limestones of the Little Dal Group Basinal assemblage, but failed to prove whether the magnetization carried by the red pigment was prefolding. The present study on "red" sandstones of the immediately underlying Mudcracked formation, using acid leaching at three sites, revealed a paleomagnetic direction *B* in the pigment (266°, −17°; seven samples; $k = 28$; $\alpha_{95} = 12°$), which closely agrees with the probable pigment direction of the Basinal assemblage (273°, −09°; 12 sites; $\alpha_{95} = 11°$). Since a fold test of the *B* sample directions is significant at the 95% probability level, I suggest that the pigment in both units is prefolding (pre-Paleocene at latest).

Une étude antérieure rapportait le détail paléomagnétique des calcaires rouges de l'assemblage Basinal, groupe Little Dal, mais ne répondait aucunement à la question soulevée de savoir si l'aimantation portée par le pigment rouge fut acquise avant l'épisode de plissement. L'étude actuelle sur les grès "rouges" de la formation de Mudcracked immédiatement sous-jacente, utilisant le lessivage acide sur trois sites, a révélé une direction paléomagnétique *B* dans le pigment (266°, −17°; sept échantillons; $k = 28$; $\alpha_{95} = 12°$), qui concorde bien avec la direction probable pour le pigment de l'assemblage Basinal (273°, −09°; 12 sites; $\alpha_{95} = 11°$). Considérant que l'épreuve de plissement sur les directions de l'échantillon *B* est significative à un niveau de confiance de 95%, je suggère que l'acquisition de l'aimantation du pigment dans les deux unités est antérieure à l'épisode de plissement (anté-Paléocène au plus tard).

[Traduit par le journal]

Can. J. Earth Sci. **21**, 371–375 (1984)

Introduction

I previously reported the paleomagnetism of red limestones of the Little Dal Group Basinal assemblage (Park 1981*a*). In this paper I present results from several sites of the oldest unit, the immediately underlying Mudcracked formation.

In the earlier paper I suggested that one of the resolved magnetizations (called *B*), thought to be carried by hematite pigment, was prefolding, and nearly coeval with the primary remanence. But evidence from a fold test was equivocal. The main purpose in publishing the present results is to prove that *B is* prefolding (pre-Paleocene at latest), and thus provide a time constraint for the acquisition of the pigment.

The present results were mainly obtained from the acid leaching of sandstones. The chemical leaching technique generally allows the direction of the removed pigment to be determined by vector subtraction or vector diagrams. The technique could not be applied in the earlier study (Park 1981*a*) of the Basinal assemblage because limestones dissolve in acid. In that study the pigment direction was obtained by a special vector technique, which requires much data for an accurate determination (Park and Roy 1979).

Geology

The Little Dal Group forms part of the Mackenzie Mountains supergroup, tentatively dated between 1200 and 770 Ma (Young *et al.* 1979). Paleontological evidence suggests an age of 1000–900 Ma for the Little Dal Group (Aitken 1981), but paleomagnetic evidence suggests either a similar age of 950–900 Ma (Park 1981*a*) or a younger age of <780 Ma (Morris and Park 1981). A radiometric date of 770 Ma from regional sills in the older Tsezotene Formation (Table 1) supports the younger age (Armstrong *et al.* 1982). These sills, which are probably related to dykes that transect the Little Dal Group as high as its Grainstone formation (Table 1), have a paleomagnetic pole (Park 1981*b*) that closely agrees with the Little Dal pole (Park 1981*a*).

The Little Dal Group comprises six formations (Aitken 1977) (Table 1). The Mudcracked formation consists mainly of

[1]Contribution from the Earth Physics Branch No. 1092.

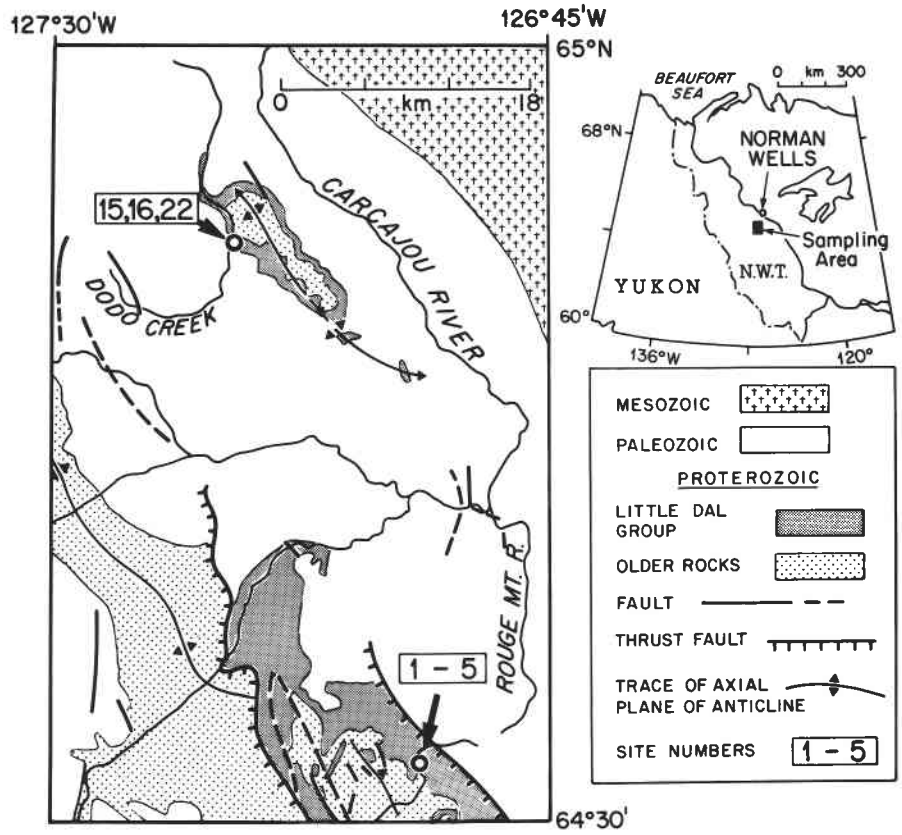


FIG. 1. Sampling localities and associated sites, together with location map.

TABLE 1. Lithostratigraphic units of the Mackenzie Mountains supergroup

	"Copper cycle"
	Coppercap Formation
	Redstone River Formation
	Basaltic lava and unnamed member
	Unconformity
Little Dal Group	
Upper	Upper Carbonate formation
	Rusty Shale formation
	Gypsum formation
Lower	Grainstone formation
	Basinal and Platform **assemblage**s, jointly **Mudcracked formation**
	Unconformity
Katherine Group	
Tsezotene Formation	
H1 unit	

NOTES: Partly after Aitken (1981). Units sampled for paleomagnetism by the author but unreported are in italics. Units reported are in boldface: Basinal assemblage (Park 1981*a*), Mudcracked formation (this paper).

sandstones and mudrocks. All but one site were collected from the sandstones. Five sites were sampled at the Rouge Mountain River locality on one limb of a general synclinal structure (Fig. 1): four were from fine maroon or gray banded sandstones over a stratigraphic thickness of about 5 m (sites 1–4), and one was from a limestone with mudstone layers, 10 m above (site 5). Two sites were sampled at the Dodo Canyon locality on the other limb, both from fine maroon banded sandstones separated by a stratigraphic thickness of 28 m (15, 16). An additional site (22) was obtained from the top of the red nodular rocks ("Dead End Shale," Aitken and Cook 1974) of the Basinal assemblage at Rouge Mountain River. It was sampled about 13 m above site 21 of Park (1981*a*).

Methods

Hand samples, collected in the field, were oriented by sun and magnetic compasses. Samples were cored in the laboratory; cores were cut into as many as five standard specimens (2.2 cm in height).

After measurement of the natural remanent magnetization (NRM), specimens were subjected to alternating field (AF) treatment, thermal demagnetization, or acid leaching, in steps. Later, the acid-leached specimens were AF or thermally demagnetized. AF and thermal treatments were conducted on apparatus described by Roy *et al.* (1969, 1972, 1973) in ambient fields generally less than 2 nT.

Acid leaching or chemical demagnetization was carried out using the general procedures of Park (1970). Thus, after axial holes had been drilled through specimens to facilitate leaching, specimens were leached in HCl (6–10 *N*), washed in water, dried, and measured. Leaching was discontinued when remanent magnetizations ceased to change with further treatment (after about 2000–7600 h in acid).

To remove any viscous remanent magnetization (VRM) acquired during leaching, several procedures were used. During the first two runs, where leaching was carried out in the Earth's magnetic field, specimens were routinely magnetically cleaned in a 5 mT AF after each leaching step, and kept in a magnetic shield for a period of time prior to measurement. During the third, and last, chemical run, leaching was carried out in a

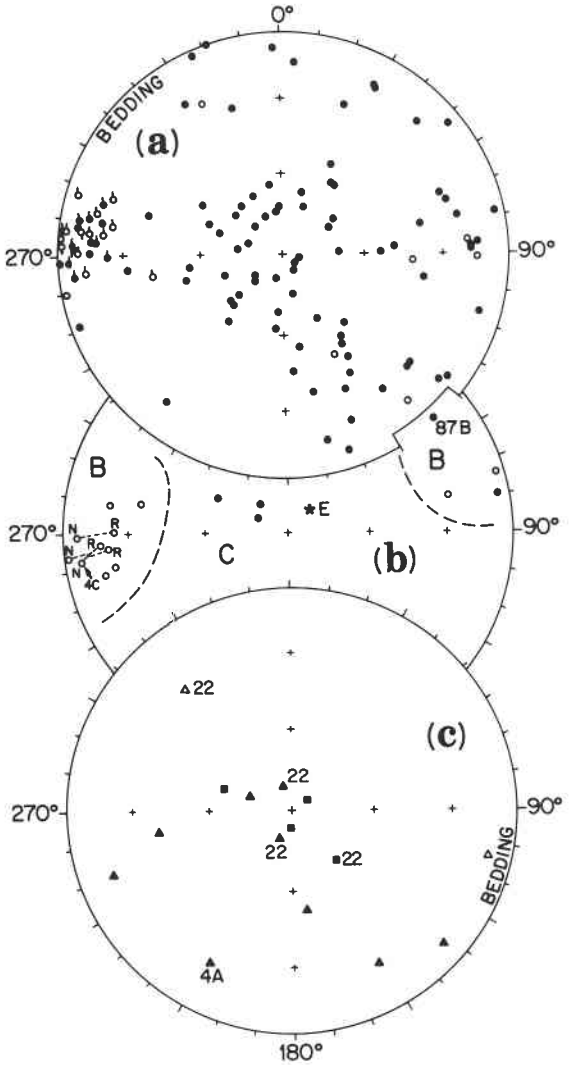


FIG. 2. Measured directions of magnetization in specimens plotted with respect to true north and either the bedding or horizontal: (*a*) natural remanent magnetization; (*b*) directions resolved using chemical leaching (circles); (*c*) directions resolved using thermal (triangles) and alternating field treatments (squares). Closed (open) symbols denote downward (upward) directed magnetizations. Labelled directions are associated with vector diagrams given in Fig. 3. In (*a*), site 1 directions are denoted by circles with attached lines. In (*b*), *B* is plotted with respect to the bedding, *C* with respect to the horizontal. E is the direction of the present Earth's field. The label 22 refers to site 22. Specimens from the same sample are joined by dashed lines. N and R indicate whether the specimen contained a normal or reverse *D* component.

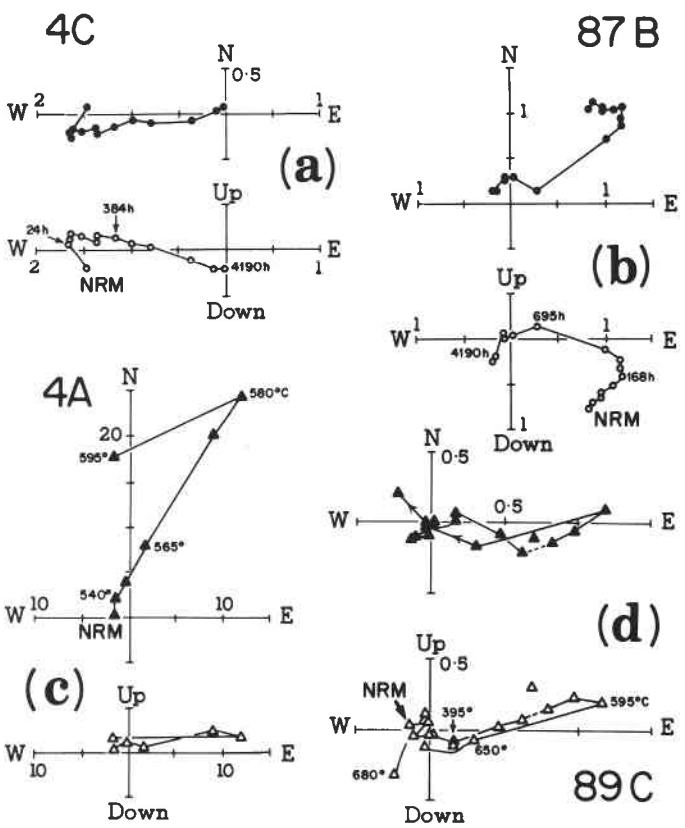


FIG. 3. Vector diagrams depicting examples of chemical leaching (circles) and thermal demagnetization (triangles) of specimens from each limb of the synclinal structure. Specimens 4C and 4A are from site 1; specimens 87B and 89C from site 15. For each specimen the vectors at each treatment step are plotted in both the horizontal (N, E) and vertical (UP, E) planes. By convention closed (open) symbols are used on the horizontal (vertical) planes; h stands for hours in acid; scales are in milliamperes per metre.

somewhat reduced field (~12% of the Earth's field). In this run specimens were not magnetically cleaned after each step, but they were kept in field-free space for a time comparable to the leaching time (Roy and Park 1974). In addition, leaching was conducted in vacuum jars under reduced pressure (about 10 kPa) (e.g., Roy and Anderson 1981), which effectively cut leaching periods in half.

Measurements were made using spinner magnetometers, either Schonstedt DSM's or a Geofyzika JR-4. The latter instrument was useful for measuring specimens with weak intensities in the range 0.02−1 mA/m.

Results and analysis

NRM directions are shown in Fig. 2*a*. Intensities generally range between 0.4 and 3.0 mA/m. Resolving the NRM proved difficult because of the generally weak intensities and the presence of multicomponents with overlapping resistive coercive force (rcf) or unblocking temperature (T_{UB}) spectra.

Most specimens were thermally demagnetized but with little success. Only 11 of 83 specimens (~13%) yielded clear-cut directions, but these were scattered (Fig. 2*c*). The T_{UB}'s, including those of site 22 from the Basinal assemblage, range up to about 580°C (e.g., 4A; Figs. 2*c*, 3*c*), consistent with a magnetite carrier. Samples from both limbs of the fold structure indicate the presence of a possible antiparallel component beyond 580°C (Fig. 3 *c* and *d*).

AF treatment of an additional 11 specimens revealed three single specimen directions from sites 5 and 16 and one from site 22 (Fig. 2*c*). All are steeply inclined normal magnetizations with rcf(s) ranging up to 25 mT.

The most successful technique for resolving the NRM was acid leaching. Of the 24 specimens put in acid, 14 or 58% yielded at least one component. Most results were garnered from site 1, which gave consistent shallowly inclined westerly directions from leaching up to 700−1100 h (Figs. 2*b*, 3*a*). A few specimens from sites 15 and 16 of the other limb of the fold structure, in addition, yielded a shallowly inclined easterly direction (Figs. 2*b*, 3*b*). These "antiparallel" directions associated with the red pigment are called *B*. During the early phase of leaching, under 300 h, a much steeper, downward-directed

TABLE 2. Directions of magnetization obtained from acid leaching

Si	Sa	Sp	Duration of treatment (h)	$D°, I°$ wrt hor	$D°, I°$ wrt bdg	R	k	$\alpha_{95}°$
				B directions				
1	5	9	264–1056	271, −32	267, −19	4.91	45	12
15	1	4	384–3015	073, −31	071, −01	—	—	—
16	1	1	24– 472	274, +02	279, −22	—	—	—
	7			269, −19	—	6.31	9	22
	7			—	266, −17	6.80	30	11
					Pole: 09°S, 143°E (δp = 6°, δm = 12°)			
				C direction				
15	3	3	<264	301, +72	262, +49	2.97	76	14

NOTES: Average site (Si) directions are given as calculated with sample (Sa) or specimen (Sp) unit weight. $D°$, $I°$: declination and inclination of the direction; R: vector resultant; k: precision; $\alpha_{95}°$: 95% angle of confidence about the average direction; h: hours in acid; δp, δm: semi-axis angles of the ellipse of confidence at $P = 0.05$ about the pole calculated from the average directions; hor: horizontal; bdg: bedding. Grand average in italics.

TABLE 3. Comparison of directions

	Location	Park (1981*a*) Si	Park (1981*a*) $D°, I°$	Park (1981*a*) $\alpha_{95}°$	Present results Si	Present results Sa	Present results $D°, I°$	Present results $\alpha_{95}°$
B	Rouge	8	275, −01	12	2	2	265, −12	—
	Dodo	5	271, −23	16	1	5	267, −19	12
	Both	13	273, −09	11	3	7	266, −17	11
C	Both	12	291, +75	5	1	3	262, +49	14

NOTE: See Table 2 for explanation of symbols.

component, *C*, was eliminated from most specimens, but it was only successfully resolved from three specimens of site 15 (Figs. 2*b*, 3*b*).

After the pigment was removed, the remanence directions were widely scattered but often with steeply inclined normal or reverse polarities, as in site 1 (e.g., after 4190 h, Fig. 3*a*). Subsequent AF or thermal treatment revealed the site 1 remanence to be composed essentially of one magnetization, a steeply inclined component *D* with rcf(s) up to about 20 mT, and T_{UB}'s up to about 500°C. However, vector diagrams displaying *D* are extremely noisy.

In individual samples, the *B* component of shallower inclination occurs in the specimen having a *D* component with normal polarity (Fig. 2*b*). This observation indicates that *D* may be biasing the *B* remanence. The effect may arise if *D* and *B* are being concurrently removed during acid leaching, that is, if detrital magnetic grains presumably containing *D* are attacked by acid before the pigment containing *B* is fully removed. Such leaching conditions may also serve to create a spurious chemical remanent magnetization (CRM) as *B* is removed, owing to the acid corrosion of the detrital grains in the presence of a constant field (Henshaw and Merrill 1980).

There is support for both of these effects. The polarity of *D* correlates with the inclination of the NRM such that where a normal *D* component is present the inclination is shallower. In addition, the younger specimen from a given core sample always yielded the normal *D* component. These observations appear to document a real component *D*, which records a magnetic reversal. But specimens were placed in constant upright or inverted positions in acid throughout the run, and these positions happen to correspond to whether either a spurious normal or reverse *D* component would be produced in the Earth's field. Since there is no corroborative evidence from the extensive previous results documenting a steep primary direction (Park 1981*a*), and the present results are few and rather noisy, I suggest that *D is* a spurious CRM.

Regardless of the nature of *D*, its effect should be reduced or largely removed when *B* data of the upright and inverted specimens are averaged. The *B* and *C* directions are given in Table 2.

Discussion and conclusions

Chemical leaching removed two magnetizations: first, a steeply inclined normal component, *C*, and then a shallowly inclined easterly or westerly component, *B*. Component *C* is only defined at site 15 from Dodo Canyon, and *B* mainly at site 1 from Rouge Mountain River.

Component *B*, whether corrected or uncorrected for bedding (Table 2), agrees well in direction with the component considered to reside in the red pigment of the Basinal assemblage (Table 3; Park 1981*a*). This agreement implied that both the results and their accuracy were probably sufficient to perform a fold test. On applying the test, the average *B* direction has a much higher precision after correction for bedding; the fold test is significant at the 95% probability level using sample unit weight (Table 4; McElhinny 1964). This test, together with the agreement between the pigment directions, implies that both the components and therefore their pigment hosts are prefolding, that is, acquired before Paleocene at the latest. Since the probable pigment direction and primary remanence direction in the Basinal assemblage closely agree, I suggest that the pigment and rock ages are nearly coeval. The *B* pole is given in Table 2.

A fold test is not possible on the *C* direction. Its relatively

TABLE 4. Application of fold test to B directions before (hor) and after (bdg) correction for bedding

	N	Average precisions (k) hor (k_1)	bdg (k_2)	k_2/k_1	95% confidence limit[a]	Significance of test
Si	3	5.95	19.30	3.25	6.39	No
Sa	7	8.74	30.00	3.43	2.69	Yes

NOTE: N is sites (Si) or samples (Sa); k is precision.
[a]Values from McElhinny (1964).

quick removal on acid leaching suggests that its carrier is more accessible to acid than is the hematite pigment and that C may be of more recent origin.

The thermal and AF results were disappointing. Directions of the resolved magnetic components were scattered. The T_{UB}'s ranging up to about 580°C suggest that magnetite could be the carrier of these components in both the Mudcracked formation and the red limestone site (22) of the Basinal assemblage. The rcf(s) were consistent with this interpretation. These directions in the Mudcracked formation could relate to detrital particles since they mainly come from unpigmented sites.

Acknowledgments

I thank J. D. Aitken of the Geological Survey of Canada for providing camp facilities, logistical support, geological advice, and student support. His summer students, Kathy Black and Kevin Sharman, kindly assisted in collecting the Little Dal sites. I am grateful to Jean Hastie for helping to measure samples and compile data. J. L. Roy and P. Lapointe of the Earth Physics Branch provided invaluable criticism. The comments of the journal reviewers, J. D. Aitken and D. Dunlop, were helpful in improving the manuscript.

AITKEN, J. D. 1977. New data on correlation of the Little Dal Formation: a revision of Proterozoic map unit 'H5'. *In* Report of activities, part A. Geological Survey of Canada, Paper 77-1A, pp. 131–135.

——— 1981. Stratigraphy and sedimentology of the upper Proterozoic Little Dal Group, Mackenzie Mountains, Northwest Territories. *In* Proterozoic basins of Canada. *Edited by* F. H. A. Campbell. Geological Survey of Canada, Paper 81-10, pp. 47–71.

AITKEN, J. D., and COOK, D. G. 1974. Carcajou Canyon map area, District of Mackenzie, Northwest Territories. Geological Survey of Canada, Paper 74-13, pp. 1–28.

ARMSTRONG, R. L., EISBACHER, G. H., and EVANS, P. D. 1982. Age and stratigraphic–tectonic significance of Proterozoic diabase sheets, Mackenzie Mountains, northwestern Canada. Canadian Journal of Earth Sciences, **19**, pp. 316–323.

HENSHAW, P. C., JR., and MERRILL, R. T. 1980. Magnetic and chemical changes in marine sediments. Reviews of Geophysics and Space Physics, **18**, pp. 483–504.

MCELHINNY, M. W. 1964. Statistical significance of the fold test in palaeomagnetism. Geophysical Journal of the Royal Astronomical Society, **8**, pp. 338–340.

MORRIS, W. A., and PARK, J. K. 1981. Correlation of upper Proterozoic strata in the Cordillera: paleomagnetism of the Tsezotene sills and the Little Dal lavas. *In* Proterozoic basins of Canada. *Edited by* F. H. A. Campbell. Geological Survey of Canada, Paper 81-10, pp. 73–78.

PARK, J. K. 1970. Acid leaching of red beds and the relative stability of the red and black magnetic components. Canadian Journal of Earth Sciences, **7**, pp. 1086–1092.

——— 1981*a*. Analysis of the multicomponent magnetization of the Little Dal Group, Mackenzie Mountains, Northwest Territories, Canada. Journal of Geophysical Research, **86**, pp. 5134–5146.

——— 1981*b*. Paleomagnetism of the Late Proterozoic sills in the Tsezotene Formation, Mackenzie Mountains, Northwest Territories, Canada. Canadian Journal of Earth Sciences, **18**, pp. 1572–1580.

PARK, J. K., and ROY, J. L. 1979. Further paleomagnetic results from the Seal Lake Group igneous rocks, Labrador. Canadian Journal of Earth Sciences, **16**, pp. 895–912.

ROY, J. L., and ANDERSON, P. 1981. An investigation of the remanence characteristics of three sedimentary units of the Silurian Mascarene Group of New Brunswick, Canada. Journal of Geophysical Research, **86**, pp. 6351–6368.

ROY, J. L., and PARK, J. K. 1974. The magnetization process of certain red beds: vector analysis of thermal and chemical results. Canadian Journal of Earth Sciences, **11**, pp. 437–476.

ROY, J. L., ROBERTSON, W. A., and KEEPING, C. 1969. Magnetic "field free" spaces for paleomagnetism, rock magnetism, and other studies. Canadian Journal of Earth Sciences, **6**, pp. 1312–1316.

ROY, J. L., SANDERS, E., and REYNOLDS, J. 1972. Un four électrique pour l'étude des propriétées magnétiques des roches. Publication de la Direction de la Physique du Globe, Canada, **42**, pp. 229–237.

ROY, J. L., REYNOLDS, J., and SANDERS, E. 1973. An alternating field demagnetizer for rock magnetism studies. Publications of the Earth Physics Branch, Canada, **44**(3), pp. 37–45.

YOUNG, G. M., JEFFERSON, G. W., DELANEY, G. D., and YEO, G. M. 1979. Middle and late Proterozoic evolution of the northern Canadian Cordillera and Shield. Geology, **7**, p. 125–128.